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Industrial communication networks – High availability automation networks – Part 4: Cross-network Redundancy Protocol (CRP)

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INDUSTRIAL COMMUNICATION NETWORKS – HIGH AVAILABILITY AUTOMATION NETWORKS –

Part 4: Cross-network Redundancy Protocol (CRP)

FOREWORD

1) The International Electrotechnical Commission (IEC) is a worldwide organization for standardization comprising all national electrotechnical committees (IEC National Committees). The object of IEC is to promote international co-operation on all questions concerning standardization in the electrical and electronic fields. To this end and in addition to other activities, IEC publishes International Standards, Technical Specifications, Technical Reports, Publicly Available Specifications (PAS) and Guides (hereafter referred to as "IEC Publication(s)"). Their preparation is entrusted to technical committees; any IEC National Committee interested in the subject dealt with may participate in this preparatory work. International, governmental and non-governmental organizations liaising with the IEC also participate in this preparation. IEC collaborates closely with the International Organization for Standardization (ISO) in accordance with conditions determined by agreement between the two organizations.
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International Standard IEC 62439-4 has been prepared by subcommittee 65C: Industrial Networks, of IEC technical committee 65: Industrial-process measurement, control and automation.

This standard cancels and replaces IEC 62439 published in 2008. This first edition constitutes a technical revision.

This edition includes the following significant technical changes with respect to IEC 62439 (2008):

– adding a calculation method for RSTP (rapid spanning tree protocol, IEEE 802.1Q),
– adding two new redundancy protocols: HSR (High-availability Seamless Redundancy) and DRP (Distributed Redundancy Protocol),
– moving former Clauses 1 to 4 (introduction, definitions, general aspects) and the Annexes (taxonomy, availability calculation) to IEC 62439-1, which serves now as a base for the other documents,
– moving Clause 5 (MRP) to IEC 62439-2 with minor editorial changes,

– moving Clause 6 (PRP) was to IEC 62439-3 with minor editorial changes,
– moving Clause 7 (CRP) was to IEC 62439-4 with minor editorial changes, and
– moving Clause 8 (BRP) was to IEC 62439-5 with minor editorial changes,
– adding a method to calculate the maximum recovery time of RSTP in a restricted configuration (ring) to IEC 62439-1 as Clause 8,
– adding specifications of the HSR (High-availability Seamless Redundancy) protocol, which shares the principles of PRP to IEC 62439-3 as Clause 5, and
– introducing the DRP protocol as IEC 62439-6.

The text of this standard is based on the following documents:

FDIS	Report on voting
65C/583/FDIS	65C/589/RVD

Full information on the voting for the approval of this standard can be found in the report on voting indicated in the above table.

This International Standard is to be read in conjunction with IEC 62439-1:2010, *Industrial communication networks – High availability automation networks – Part 1: General concepts and calculation methods.*

A list of the IEC 62439 series can be found, under the general title *Industrial communication networks – High availability automation networks*, on the IEC website.

This publication has been drafted in accordance with ISO/IEC Directives, Part 2.

The committee has decided that the contents of this amendment and the base publication will remain unchanged until the stability date indicated on the IEC web site under "http://webstore.iec.ch" in the data related to the specific publication. At this date, the publication will be

• reconfirmed,
• withdrawn,
• replaced by a revised edition, or
• amended.

A bilingual version of this standard may be issued at a later date.

INTRODUCTION

The IEC 62439 series specifies relevant principles for high availability networks that meet the requirements for industrial automation networks.

In the fault-free state of the network, the protocols of the IEC 62439 series provide ISO/IEC 8802-3 (IEEE 802.3) compatible, reliable data communication, and preserve determinism of real-time data communication. In cases of fault, removal, and insertion of a component, they provide deterministic recovery times.

These protocols retain fully the typical Ethernet communication capabilities as used in the office world, so that the software involved remains applicable.

The market is in need of several network solutions, each with different performance characteristics and functional capabilities, matching diverse application requirements. These solutions support different redundancy topologies and mechanisms which are introduced in IEC 62439-1 and specified in the other Parts of the IEC 62439 series. IEC 62439-1 also distinguishes between the different solutions, giving guidance to the user.

The IEC 62439 series follows the general structure and terms of IEC 61158 series.

The International Electrotechnical Commission (IEC) draws attention to the fact that it is claimed that compliance with this document may involve the use of a patent concerning a full-duplex Ethernet in which each device periodically transmits a message representing its connectivity to the other devices , allowing them to choose a redundant path in case of failure, given in 7.1 and 7.3.

IEC takes no position concerning the evidence, validity and scope of this patent right.

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Austin, TX 78759

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INDUSTRIAL COMMUNICATION NETWORKS – HIGH AVAILABILITY AUTOMATION NETWORKS –

Part 4: Cross-network Redundancy Protocol (CRP)

1 Scope

The IEC 62439 series is applicable to high-availability automation networks based on the ISO/IEC 8802-3 (IEEE 802.3) (Ethernet) technology.

This part of the IEC 62439 series specifies a redundancy protocol that is based on the duplication of the network, the redundancy protocol being executed within the end nodes, as opposed to a redundancy protocol built in the switches. The switchover decision is taken in each node individually. The cross-network connection capability enables single attached end nodes to be connected on either of the two networks.

2 Normative references

The following referenced documents are indispensable for the application of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

IEC 60050-191, *International Electrotechnical Vocabulary – Chapter 191: Dependability and quality of service*

IEC 62439-1:2010, *Industrial communication networks – High availability automation networks – Part 1: General concepts and calculation methods*

ISO/IEC 8802-3:2000, *Information technology – Telecommunications and information exchange between systems – Local and metropolitan area networks – Specific requirements – Part 3: Carrier sense multiple access with collision detection (CSMA/CD) access method and physical layer specifications*

3 Terms, definitions, abbreviations, acronyms, and conventions

3.1 Terms and definitions

For the purposes of this document, the terms and definitions given in IEC 60050-191, as well as in IEC 62439-1, apply.

3.2 Abbreviations and acronyms

For the purposes of this document, the abbreviations and acronyms given in IEC 62439-1, apply, in addition to the following:

DANC Doubly attached node implementing CRP

SANC Singly attached node implementing CRP

3.3 Conventions

This document follows the conventions defined in IEC 62439-1.

4 CRP overview

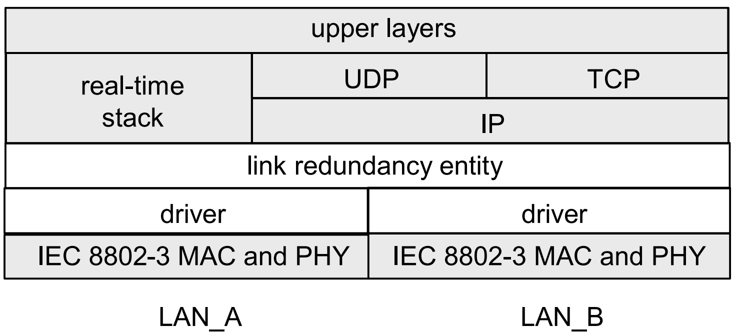
This International Standard specifies a redundancy protocol that is based on the duplication of the network, the redundancy protocol being executed within the end nodes, as opposed to a redundancy protocol built in the switches. There is no central “redundancy manager”; instead each node operates autonomously. The cross-network connection capability enables single attached end nodes to be connected on either of the two networks.

5 CRP nodes

There exists different classes of nodes that may interoperate on the same network:

- DANCs (Doubly Attached Nodes) able to execute the CRP protocol, and having two ports for the purpose of redundancy.
- SANCs (Singly Attached Nodes) able to execute the CRP protocol, and having only one port.
- SAN (Singly Attached Nodes), such as commercially available laptops or file servers that are not aware of the CRP protocol. Even though not aware, SANs can also have access to the redundancy management data for the purpose of monitoring and network management.

In DANCs, these two ports are referred to as port A and port B. They are managed by the Link Redundancy Entity (LRE), whose implementation is not prescribed, and which is conceptually located in the communication stack below the network layer, as illustrated in Figure 1.



IEC 380/10

Figure 1 – CRP stack architecture

This arrangement provides application-level transparency. The LRE hides redundancy from the upper layers and manages the ports. A node can therefore operate with only one IP address.

6 CRP LAN topology

Implementing the redundancy protocol within the DANCs allows a variety of topologies, using switches that are not aware of the redundancy protocol and could implement another redundancy protocol such as RSTP.

This International Standard does not dictate the topology, but does allow for configuration of node behaviour to accommodate the characteristics of the specific LAN being used.

Nodes may be attached to the same or to different switches of a single LAN, which may or may not include redundant links, as Figure 2 shows. Attaching both links to the same switch only provides leaf link failure resilience.

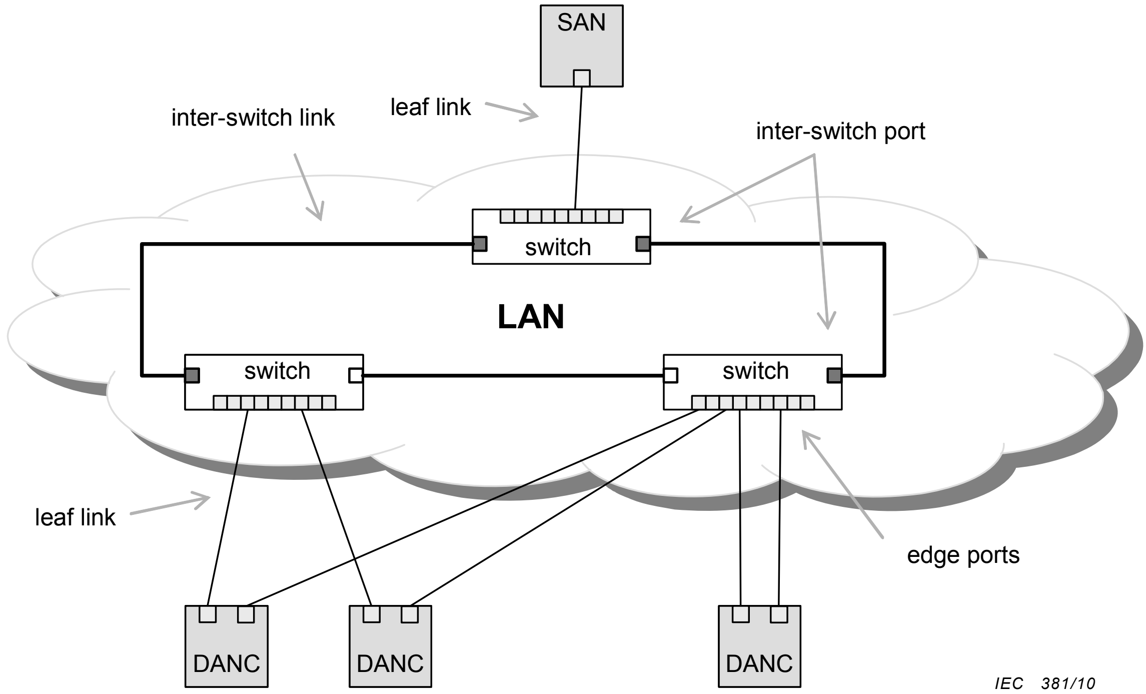


Figure 2 – CRP single LAN topography

Nodes may be attached to separate LANs, which are basically failure-independent, but may be connected by an inter-LAN link, as Figure 3 shows.

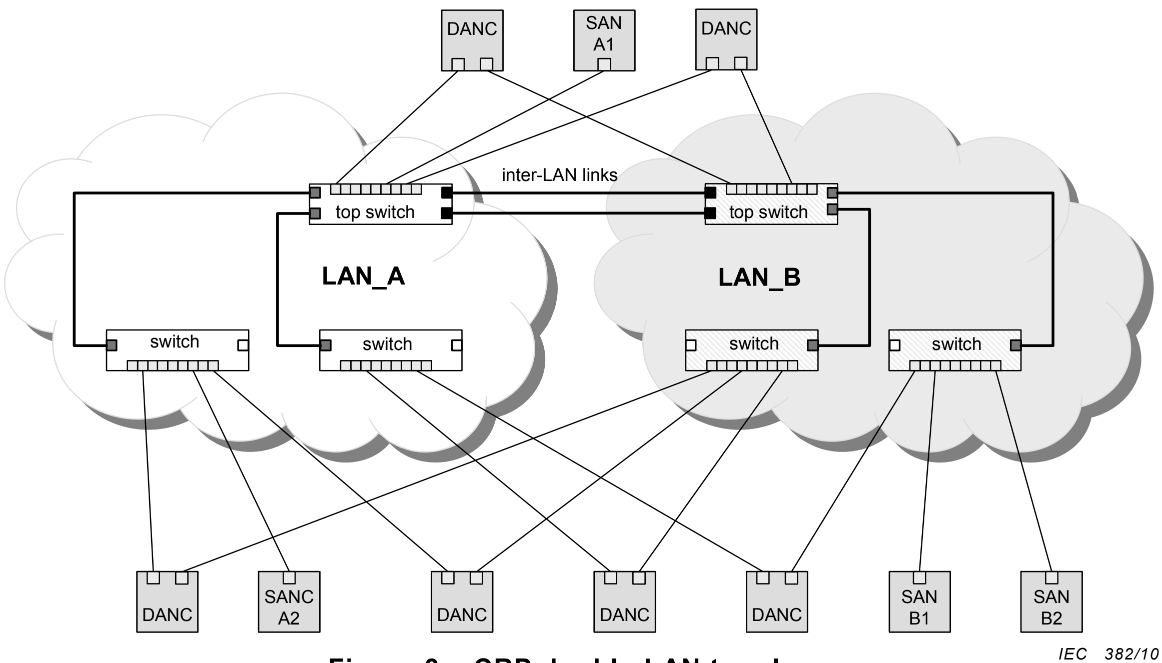


Figure 3 – CRP double LAN topology

When there is only one LAN, a node is attached through both its ports to that LAN. In double LAN configurations, port A is normally connected to LAN_A and port B to LAN_B. Connecting a node twice to the same network tree or connecting port A to LAN_B and vice-versa may be a configuration error called "crossed cables".

7 CRP key components

7.1 CRP general protocol operation

7.1.1 Doubly-attached nodes (DANCs)

DiagnosticFrames are used to exercise communication paths and to assess the network health. A DiagnosticFrame contains a summary of the reporting node's view of the network health and status, including its own port.

Annunciation frames are sent to announce the existence of the node. These frames are described in 8.7.1

Each DANC sends a pair of DiagnosticFrames periodically, every T_{dmi}, on both of its ports, as Figure 4 shows. Each DANC that receives one DiagnosticFrame on one port expects the other message of the pair on the other port. (On a single LAN, the node receives both messages on both ports.) If a node receives no message or if it does not receive the second DiagnosticFrame on the other port before receiving several more DiagnosticFrames on the same port, it records a fault in the row of the Network_Status_Table for the corresponding node.

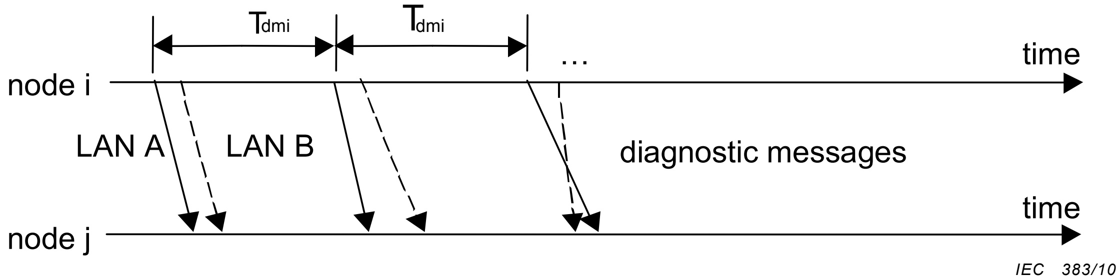


Figure 4 – CRP DiagnosticFrame pair approach

In practice the receiving node compares the Sequence_Number of the last message received on the other port with that just received. If the difference in Sequence_Number is more than the configured Max_Sequence_Number_Difference, a fault is recorded.

Based upon the diagnostic frames it receives from all other nodes, each node can select which port to use to send messages to a particular node, on a node-per-node basis.

EXAMPLE Figure 5 shows four nodes connected to two redundant LANs which are not connected with each other. Node 3 and node 4 have link failures. The diagnostic frame handling on node 3 is detailed.

Each node broadcasts its view on the port status of all nodes it detected in addition to other status information (source MAC address, Node_Index, etc.).

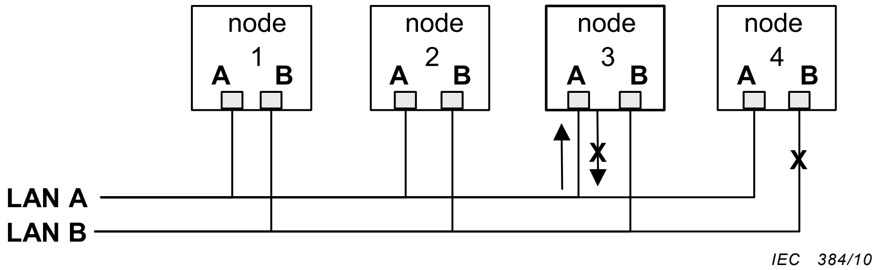
Node 3 maintains a Network_Status_Table populated by the DiagnosticFrames from nodes 1, 2, and 4, as shown in Table 1.

The port status values are OK to indicate a working condition, and X for a don't know or bad condition.

According to the first three columns of the Network_Status_Table in Table 1, node 3 sends out its Received_DiagnosticFrame for port A as [OK, OK, OK, OK] and for port B as [OK, OK, OK, X].

Similarly, node 1 sends out its view on nodes 2, 3, and 4 as [OK, OK, X, OK] for port A and [OK, OK, OK, X] for port B. Node 3's adapter A and adapter B status is populated as shown in Table 1.

The row for node 3 is set based on its own testing, but in this example there is no testing, so all appears to be OK.



Node 3: Interface A partial failure; can receive, but not transmit
Node 4: Interface B complete failure.

Figure 5 – CRP example system

Table 1 – CRP example Network_Status_Table for node 3

Node_Index Number	Received_DiagnosticFrame		Reported status extracted from DiagnosticFrame	
	Received on adapter A	Received on adapter B	Node 3 Received on adapter A	Node 3 Received on adapter B
	Received from adapter A/B [a]	Received from adapter A [a]/B	Received from adapter A/B [a]	Received from adapter A [a]/B
1	OK/X	X/OK	X/X	X/OK
2	OK/X	X/OK	X/X	X/OK
3 (this node)	OK/X	X/OK	OK/X	X/OK
4	OK/X	X/X	X/X	X/X

[a] The cross statuses are all “X” for a dual LAN without inter-LAN link. That is, messages originating from a port A are never heard on a port B and vice versa.

The DiagnosticFrames provide therefore:

- minimal assurance of a working path. With each message received, the receiving node can assume that its own receiver, the reporting node’s transmitter, and the path through the network are all working;
- assurance that the reverse path is working. With each message received, the receiving node can extract the reporting node’s view of the receiving node and thus determine whether its own transmitter, the reporting node’s receiver, and the path through the network are all working.

This allows the system administrator to construct a variety of coverage strategies such as:

- ensure that all paths between all nodes are tested;
- send to a single node. This node may be a “diagnostic node” that only provides detection of faults between each node and the diagnostic node.

7.1.2 Singly attached nodes

Singly attached nodes (SANCs) also can send and receive DiagnosticFrames. If they choose to transmit them, the DANCs and SANCs are aware of their presence and attempt to ensure that messages reach them. The Network_Status_Table built by the SANC allows it to build DiagnosticFrames and also, in a single LAN, to select a path to a node with a failed port.

7.2 CRP statistics

Statistics should be gathered and presented for each port, by a system management application. Examples of presentation methods are a graphical user interface for visual reporting of network errors or via SNMP.

7.3 CRP Network_Status_Table

Each node maintains a Network_Status_Table that holds the node's view of the network.

This table is used to assist with selection of which port(s) to use for transmission to a destination address and which port(s) to use for reception of multicast transmissions.

The Network_Status_Table is constructed from received DiagnosticFrames as well as from other locally acquired and sometimes vendor-specific diagnostic information, for example built-in tests, link integrity pulse, etc.

This table is conceptual and is described to assist with understanding of the concepts in this specification and no specific implementation is prescribed or implied. It is therefore not visible to network management; however, the contents of the table are reflected in the DiagnosticFrames.

The Network_Status_Table in each node keeps for each node, and in some cases for each port of each node the following information:

a) for each remote reporting node,
   - remote node identification (name, index in a table, etc.);
   - diagnosis message interval;
   - apparent number of ports (1..N);
   - for each port, in nominal order (e.g., port to LAN_A before port to LAN_B) the MAC address of the remote port.
b) for each pairing of local and remote ports (e.g., A/B, B/A, A/A or B/B)
   - time of latest message receipt;
   - sequence number of latest received message;
   - receipt of DiagnosticFrame from remote port within time;
   - remote port's link status from last received DiagnosticFrame.
c) for assessment of remote connectivity
   - inferred status of set of ports (functioning properly, cross-connected, ...);
   - preferred port pairing.

EXAMPLE Table 2 shows a Network_Status_Table for a SANC and Table 3 a Network_Status_Table for a DANC.

Table 2 – CRP Network_Status_Table for singly connected nodes

Reporting node information			Messages received here sent from reporting node's adapter A				Messages received here sent from reporting node's adapter B			
Node_Index and Node_Name	Number of adapters	Interval (ms)	Time	Sequence_ number of last message received	Diagnostic Frame received	Reported status extracted from DiagnosticFrame for AA	Time	Sequence_ number of Last message received	Diagnostic Frame received	Reported status extracted from Diagnostic Frame for BA
1 FD001	1	2 000	Time	Number	OK	OK				
2 FD002	1	2 000	Time	Number	OK	OK				
3 LD001	1	1 000					Time	Number	OK	OK
4										
5 LD003	1	5 000	Time	Number	OK	OK				
6	2	5 000	Time	Number	OK	OK	Time	Number	OK	OK
7	1	5 000	Time	Number	OK	OK				

Table 3 – CRP Network_Status_Table for DANC

Reporting node information				Messages received here on adapter A					B[a]		Assessment for reporting node	
				A[b]	Sent from reporting node's adapter B				A[c]	B[d]		
Node_Index and Node_Name	**Addresses for adapters A, B**	**Number of adapters**	**Interval (ms)**	**…**	**Time**	**Sequence_ Number of last message received**	**Diagnostic Frame received**	**Reported status extracted from Diagnostic Frame for BA**	**…**	**…**	**Crossed_ cable_status**	**Selected transmission adapter and receiving adapter**
1 FD001	addr1.A, addr1.B	2	2 000	…	Time	Number	OK	OK	…	…	OK	BB
2 FD002	addr2.A, addr2.B	2	2 000	…	Time	Number	OK	OK	…	…	OK	AA
3 LD001	addr3.A, addr3.B	2	1 000	…	Time	Number	OK	OK	…	…	OK	AA
4												
5 LD003	addr5.A, addr5.B	2	5 000	…	Time	Number	OK	OK	…	…	OK	AA
6	addr6.A, addr6.B	2	5 000	…	Time	Number	OK	OK	…	…	OK	BB
7	addr7.A	1	5 000	OK	N/A	N/A	N/A	N/A	NA	NA	OK	AA

a Messages received here on adapter B.

b Sent from reporting node's adapter A.

c Sent from reporting node's adapter A.

d Sent from reporting node's adapter B.

7.4 CRP recovery time

7.4.1 Recovery time calculation

The maximum recovery time from a fault is:

$$t_{rec} = (1+\text{Max\_Sequence\_Number\_Difference}) \times t_{dmi} + t_{path} + t_{proc}$$

where

t_{rec} is the recovery time;

t_{dmi} is the time interval of diagnostic frames;

t_{path} is the latency of frame delivery of the network; and

t_{proc} is the processing time of the receiving LAN redundancy entity.

The value of t_{path} is determined by the amount of time the packet takes to travel through the network. In a switched network the worst case transition time through a FIFO based switch is dependent on the number of ports of the switch. It may be necessary to use QoS to respect this delay. The arrival of the packet of interest after all other packets bound for the destination interface of interest produces the worst case delay. In this case the delay is:

$$t_{sw} = N_{ports} \times t_{rate} \times S_{pak} \times 8$$

where

t_{sw} is the delay time in a single switch;

N_{ports} is the number of switch ports on a single switch;

t_{rate} is the 1/data rate; and

S_{pak} is the max packet size in bytes.

EXAMPLE The following are examples of recovery time for various end node speeds.

Processing time is dominated by the interrupt response time of the end node. For this example an interrupt time of 15 µs is used.

Given: t_{dmi} = 400 ms, t_{proc} = 15 µs, Max_Sequence_Number_Difference = 1, N_{ports} = 24 and S_{pak} = 1 522 bytes for all cases.

For all end nodes with 10 Mbit/s bandwidth:

$$t_{sw} = 24 \times 10^{-7} \times 1\,522 \times 8 = 0{,}029 \text{ s}$$

In a single redundant LAN using 6 switches of 24 ports each, and assuming the diagnostic packet traverses all switches, the total t_{path} is $6 \times t_{sw}$ = 0,174 s.

Thus,

$$t_{rec} = 2 \times 0{,}40 \quad + 0{,}174 + 0{,}000\,015 \text{ s}$$

For this example the maximum processing time is negligible thus:

$$t_{rec} = 0{,}974 \text{ s}$$

For all end nodes with 100 Mbps bandwidth, the equation scales directly thus

$$t_{sw} = 24 \times 10^{-8} \times 1\,522 \times 8 = 0{,}002\,9 \text{ s and for 6 paths} = 0{,}014\,6 \text{ s}$$

Thus

$$t_{rec} = 2 \times 0{,}40 + 0{,}017\,4 + 0{,}000\,015$$

$$t_{rec} = 0{,}817\,4 \text{ s}$$

7.4.2 Maximum repair time

For faults such as cable breaks there is no repair time. However, in a multiple interface switch topology repairing a failed interface requires replacing the entire switch. In this case powering the switch down to replace it causes additional network disruptions in nodes that have paths though that switch. So as a worst case, the repair time is identical to the recovery time from a fault described in 7.4.1.

7.5 CRP multicast messages

7.5.1 Sending

Multicast messages are always sent from both ports (if both are operational). They carry as source MAC address the address of the port over which they have been sent.

This applies in particular to the DiagnosticFrames and AnnunciationFrames

7.5.2 Receiving

On some network topologies, a single operational multicast message can be received on each of a node's ports. If a node has two ports, the node may be configured to use the Network_Status_Table to select a reception port for multicast operational messages, and so reduce its interrupt and message processing load. Duplicates still need to be detected and discarded.

7.6 CRP unicast messages

7.6.1 Sending a frame

Each unicast frame sent by a CRP redundancy participating node is sent from only one port.

When the LRE receives a frame from the IP stack (or other upper layer), it examines the frame, identifies the destination MAC address and looks up for that address in the Network_Status_Table.

If the A-A path to the destination is OK in the table, the LRE sends the frame to port A, which inserts its address as a source MAC address.

If A-A path is NOT OK in the table, but the A-B path is OK, then the LRE substitutes the B MAC address of the destination node in the frame and sends the frame to port A, which inserts its address as a source MAC address.

If the A-A and A-B paths are both NOT OK, but the B-A path is OK, the LRE sends the frame to port B, which inserts its address as a source MAC address.

If the A-A, A-B and B-A paths are NOT OK, the LRE substitutes the B MAC address of the destination node in the frame and send the frame to port B, which inserts its address as a source MAC address.

If the message is broadcast or multicast, the LRE sends the frame to A if A is working and to B if A is not working.

7.6.2 Receiving a frame

When the LRE receives a frame from the physical layer over port B, it substitutes the destination MAC B to MAC A address, before forwarding the frame up to the stack. Some IP stacks are sensitive to the IP/MAC association being correct. The IP address is not manipulated/substituted in any way on neither transmit nor receive.

7.7 CRP redundancy information

Each node is configured by a user definable mechanism. The configuration determines the details of how each node transmits DiagnosticFrames and how it uses the Network_Status_Table to select transmission and reception ports. This information can be obtained from another node by listening to AnnunciationFrames.

7.8 CRP redundancy statistics

Any node (even if it is not CRP enabled) may gather and display the health of the nodes in the network by subscribing to the multicast address used for the DiagnosticFrames. The application may then generate a Network_Status_Table and use it in any way.

8 CRP protocol

8.1 CRP singly attached node

SANCs shall have one port and shall participate in the CRP redundancy protocol.

8.2 CRP doubly attached node

DANCs shall have two ports and participate in the CRP redundancy protocol.

8.3 CRP Installation, configuration and repair

DANC shall be connected to a single redundant LAN with redundant leaves or to a redundant LAN without redundant leaves as described in Clause 6.

NOTE 1 The former connection provides four possible paths to other doubly connected nodes while the latter provides only two.

In order to achieve switch and leaf link redundancy, each port of a DANC shall be connected to a different switch.

SANC and SAN may be connected to any LAN, but preferably all SANCs and SAN shall be connected to the same LAN.

The assigned Node_Index and Node_Name shall be unique in the network.

The maximum Node_Index is 2 048; the value of 0 shall not be used for Node_Index.

The maximum number of nodes is 2 047.

NOTE 2 This number is limited by the size of the array of status available in the diagnostic packet.

The maximum number of network switch layers of a single redundant LAN with redundant leaves shall be 3.

NOTE 3 This limitation maintains a spanning tree diameter of 7 hops for those networks where spanning tree is used to prevent loops.

8.4 CRP LRE model attributes

8.4.1 Attribute specification

8.4.1.1 Protocol_Version

This configured attribute specifies the CRP protocol used. It is an Unsigned8 with a value of 0x01.

Packets with higher version numbers shall be rejected by lower versions. Newer versions shall revert to compatibility mode when older version packets are received.

8.4.1.2 Number_of_ports

This configured attribute specifies the number of ports on this node for the purpose of redundancy (Unsigned8 = 1 or 2).

8.4.1.3 Max_Sequence_Number_Difference

This configured attribute specifies the maximum acceptable difference between the Sequence_Number parameters in a pair of DiagnosticFrame received from a particular sending node. It shall be an Unsigned8 greater than or equal to 1.

NOTE This attribute affects the speed and accuracy of the dual message approach. The value of Max_Sequence_Number_Difference number is at least one to ensure that faults are not detected by normal incrementing of the Sequence_Number. A number of at least two ensures that a single lost message does not cause detection of faults. Having larger numbers allows tolerance of the loss of several successive messages but slows down speed with which the algorithm detects actual faults, as a trade-off between transient tolerance and detection speed.

8.4.1.4 Redundancy_Flags

This configured attribute specifies five flags that are not transmitted, indicating one or more of the following:

a) SingleMulticastMessageTransmissionEnabled. This defines the transmission policy for all services with multicast destination addresses except for the DiagnosticFrame.
   - False Transmit on both ports
   - True Transmit on one port

b) CrossedCableDetectionEnabled.
   - False Do not detect crossed cables
   - True Detect crossed cables

c) SinglePortMulticastMessageReceptionEnabled. This defines the reception policy for all multicast frames except for the DiagnosticFrame_Addresses for ports A and B.
   - False Listen for multicast addresses on both ports
   - True Listen for multicast addresses on one port except if a fault is detected

d) DiagnosisUsingOwnMessagesEnabled.
   - False Do not use own DiagnosticFrames for diagnosis
   - True Use own DiagnosticFrames for diagnosis

e) LoadBalancingEnabled.
   - False Do not balance load
   - True Balance load

8.4.1.5 DiagnosticFrame_Interval

This configured attribute specifies the time interval in milliseconds between successive sending of the DiagnosticFrames as an Unsigned32.

NOTE This parameter is set individually for each end device. This allows tuning of the intervals to provide fast detection of critical end devices while reducing the traffic from other end devices.

8.4.1.6 Ageing time

This configured attribute specifies the time interval in milliseconds used by the LRE to remove silent nodes from its Network_Status_Table.

NOTE 1 The configuration ensures that the value of this attribute is larger than the value of the Max_Sequence_Number_Difference attribute multiplied by the largest value of the DiagnosticFrame_Interval attribute found in received DiagnosticFrames.

NOTE 2 Short ageing times (minutes) mean that DANCs appear and disappear if their power fails briefly. Longer ageing times (days) mean that DANCs that have been removed continue to appear in the Network_Status_Tables.

8.4.1.7 DiagnosticFrame_Address

This configured attribute specifies the 32-bit IP address for DiagnosticFrames. The address may be a broadcast or multicast IP address.

The address 224.0.0.105 has been registered for this purpose.

8.4.1.8 DiagnosticFrame_UDP_source_port

This configured attribute specifies the 16-bit UDP source port used for DiagnosticFrames.

The same port number as for AnnunciationFrames shall not be used.

8.4.1.9 DiagnosticFrame_UDP_destination_port

This configured attribute specifies the 16-bit UDP destination port used for DiagnosticFrames.

The same port number as for AnnunciationFrames shall not be used.

8.4.1.10 Annunciation_UPD_port

This fixed attribute specifies the 16-bit UDP destination source and destination port used for AnnunciationFrames as an Unsigned16 with a value of 1 089.

8.4.1.11 Node_Index

This configured attribute specifies the 16-bit Node_Index of that node.

NOTE The Node_Index is unique for each node in the scope of the DiagnosticFrame_Address.

8.4.1.12 Max_Node_Index

This configured attribute specifies the highest expected Node_Index.

NOTE This number is used to determine the length of the adapter status array and the crossed cable array in the DiagnosticFrame as well as the length of the Network_Status_Table in each end device within the network. The value of this parameter affects the size of DiagnosticFrames; therefore having a very large number adversely impacts performance.

It is recommended that the Max_Node_Index be set to the same value for all devices within the network.

8.4.1.13 Node_Name

This configured attribute is a VisibleString32 octets that specifies a Node_Name that is unique for each node in the scope of the DiagnosticFrame_Address. Unused octets in this field shall be transmitted as ASCII space characters.

8.4.1.14 Annunciation_Interval

This configured attribute specifies the time interval at which AnnunciationFrames are sent.

8.4.1.15 Operational_IP_Address

This configured attribute specifies the IP address used for application communication.

8.4.1.16 Duplicate_Detection_State

This dynamic attribute indicates possible detections of multiple addresses.

It shall be reset to all 0 by configuration or start-up and set upon detection of a conflicting network situation.

It is a bit set encoded as:

0	1	2	3	4	5	6	7
RES						DND	DID

Bit 0 — 5: Reserved
set to 0

Bit 1: DND
1: duplicate name detected,

Bit 0: DID
1: duplicate node index detected,

8.4.1.17 LRE state

This dynamic attribute specifies the state of the LRE as one octet.

0	1	2	3	4	5	6	7
CONF							SYN

Bit 0 — 6: CONF
0 = reserved
1 = no name configured
2 = operational
3 – 127 = reserved

Bit 7: SYN
0 = Not synchronized with time server
1 = Synchronized with time server

The LRE state shall be reset to all “0” by configuration or start-up.

8.4.1.18 Sequence_Number

This dynamic attribute is a monotonically increasing 32-bit integer that shall start with 0 for the first DiagnosticFrame sent after start-up, incrementing by 1, and rolling over to 0.

8.4.1.19 Path_Status

This dynamic attribute summarizes the view of the node on its paths. It consists of four Path_Status sets as Table 4 shows.

Table 4 – CRP Path_Status_Sets

	Remote node sent over	This node received over
Path_Status_A_to_A	A	A
Path_Status_B_to_A	B	A
Path_Status_A_to_B	A	B
Path_Status_B_to_B	B	B

Each Path_Status set consists of a 32-bit aligned sequence of bits, each bit representing a node, using the Node_Index as an offset and indicating the status of the transmission path from the reporting node to this node.

The following values are used for each bit:

- 0 = DiagnosticFrames successfully received
- 1 = DiagnosticFrames not received

EXAMPLE Table 5 shows a Path_Status set for the node with Node_Index 4. This node (4) indicates that it sees:

Node 1: received “A” messages over its LAN_A, and “B” messages over its LAN_B.

Node 2: did not receive messages for a time longer than Aging_Time.

Node 3: received only “A” messages over its LAN_A (it’s possibly a SAN).

Node 4: received its own messages over the same port they were sent.

Node 5: received “A” messages over its LAN_A, and “B” messages over its LAN_A – possible configuration error.

Node 6: received “A” messages over its LAN_B, and “B” messages over its LAN_A – possible crossed cable error.

Table 5 – CRP example of a Path_Status_Set

0		2		4		6		8		10		12		14		16		18		20		22		24		26		28		30	
Path_Status_A_to_A																															
0	1	0	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Path_Status_B_to_A																															
0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Path_Status_A_to_B																															
0	0	0	0	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Path_Status_B_to_B																															
0	1	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

8.4.1.20 Configuration_Version

This attribute specifies the version of this object as an Unsigned16. Each time the value of any of its other attributes changes by configuration, the version number shall be incremented by 1. Version number 0 indicates that this object has not been configured. This value shall be skipped when rolling over.

8.4.2 Impact of LRE configuration attributes

The impact of the LRE configured attributes is summarized in Table 6 .

Table 6 – CRP configuration attributes impact on LAN operation

Configuration parameter	Dual LAN	Single LAN
Node_Index	Same for both, see 8.4.1.11	
Max_Node_Index	Same for both, see 8.4.1.12	
Max_Sequence_Number_Difference	Same for both, see 8.4.1.3	
DiagnosticFrame_UDP_destination_port	Same for both, see 8.4.1.9	
Redundancy_Flags	This parameter consists of the following five flags	
SingleMulticastMessageTransmissionEnabled	Normally false, allowing multicast messages to be sent on both LANs if the error conditions and presence of SANCs warrants it. Where receiving node loading is critical, however, this may be set true	Normally set true, allowing multicast messages to propagate to both adapters of other nodes. Sending on both adapters increases traffic on the LAN and the receiving nodes
CrossedCableDetectionEnabled	Normally set true. This allows detection of crossed cables	Normally false. Crossed cables have irrelevant on a single LAN
SingleMulticastMessageReceptionEnabled	Normally false, allowing reception of multicasts on either adapter. Because each multicasting node can choose which to send on, listening on both adapters is essential to hear them	Normally true, allowing operating multicast messages propagate to both adapters of other nodes. Listening on both adapters increases traffic on the receiving node
DiagnosisUsingOwnMessagesEnabled	Normally false, preventing DiagnosticFrame sent on one adapter will not be heard on the other	Normally true, allowing DiagnosticFrames sent from one adapter to be received on the other. Using this as a diagnostic improves fault detection, at the cost of processor load
LoadBalancingEnabled	May be set false on a temporary basis to facilitate system maintenance	Normally set false. Load balancing is not applicable to single LAN networks
DiagnosticFrame_Interval	Same for both, see 8.4.1.5	
Aging_Time	Same for both, see 8.4.1.6	
DiagnosticFrame adapter A send address DiagnosticFrame adapter B send address	The address to which DiagnosticFrames are sent on adapters A and B. Usually the same address is used for both adapters	The addresses to which DiagnosticFrames are sent on adapters A and B. The same address may be used to maximize the ability to detect and recover from faults, at a cost of CPU load. Different addresses allow a reduction of traffic seen by the receiving node
DiagnosticFrame adapter A receive address	The address to which DiagnosticFrames are received on adapters A and B. Usually the same address is used for	The addresses to which DiagnosticFrames are received on adapters A and B. The same address may be

Configuration parameter	Dual LAN	Single LAN
	both adapters	used to maximize the ability to detect and recover from faults, at a cost of CPU load

8.5 CRP encoding of the DiagnosticFrame

A DiagnosticFrame shall have the content described in Table 7.

Table 7 – CRP DiagnosticFrame format

Parameter name	Offset	Data type	Size	Description
Ethernet DLL header				
Preamble	0		8	Alternating ones and zeros
Destination address	8		6	Broadcast, multicast MAC address
Source address	14		6	Unicast source MAC address of adapter used to send this message.
Type	20		2	0x800 for IP datagrams
IP header				
IP version	22		4	IP version (4 bit field) = 4
IP header length				Internet header length (4 bit field) is the length of the internet header in 32 bit words = 6
IP type of service				Type of service (8 bit field) – set all fields = 0 Bits 0-2: precedence. Bit 3: 0 = normal delay, 1 = low delay. Bits 4: 0 = normal throughput, 1 = high throughput. Bits 5: 0 = normal reliability, 1 = high reliability. Bit 6-7: reserved for future use.
IP total length				Length of IP field (16 bit field) in bytes, including IP header and data = 346
IP identifier	26		4	Identifier for fragmented packets. Set = 0
IP flags				Flags (3 bit field) Bit 0: reserved, shall be zero Bit 1: don't fragment = 1 Bit 2: 0 = last fragment
IP fragment offset				Position in original datagram. First fragment = 0
IP time to live	30	Unsigned8	4	Set to 1
IP protocol		Unsigned8		User datagram = 17 (decimal)
IP header checksum				16 bits. The checksum field is the 16 bit one's complement of the one's complement sum of all 16 bit words in the header. For purposes of computing the checksum, the value of the checksum field is zero.
IP source address	34		4	Source_IP_Address.
IP destination address	38		4	Configured DiagnosticFrame_Address, see 8.4.1.7
IP options	42		1	
IP pad	43		3	Pad header to a 32 bit boundary. Pad set = 0
UDP header				
UDP source port	46		2	See 8.4.1.8
UDP destination port	48		2	See 8.4.1.9
UDP length	50		2	Length of UDP field
UDP checksum	52		2	Checksum is the 16-bit one's complement of the one's complement sum of a pseudo header of information from the IP header, the UDP header, and the data.
CRP header				
CRP Protocol_Version	54	Unsigned8	1	see 8.4.1.1
reserved	58	Unsigned24	3	Set to 0x801001
CRP auxiliary address	58	Unsigned32	4	Not used – set to 0
CRP message length	62	Unsigned32	4	Specifies the number of octets contained in the entire message, starting immediately after the UDP header until the end of the message.

Parameter name	Offset	Data type	Size	Description
CRP Body				
CRP Node_Index	66	Unsigned16	2	See 8.4.1.11
CRP Number_of_ports	68	Unsigned8	1	See 8.4.1.2
Transmission adapter	69	Unsigned8	1	Port used to transmit this DiagnosticFrame. 0 = adapter A 1 = adapter B
CRP DiagnosticFrame_Interval	70	Unsigned 32	4	See 8.4.1.5
CRP Node_Name	74	VisibleString	32	See 8.4.1.13
Reserved	106	Unsigned8	1	Reserved, set to zero
CRP Duplicate_Detection_State	107	Unsigned8	1	See 8.4.1.16
Number_Of_Adapter_Statuses	108	Unsigned16	2	Number of Unsigned32 entries in the Path_Status.
Path_Status_A_to_A	110	array of Unsigned32	See [a]	See Table 4
Path_Status_B_to_A	See a)	array of Unsigned32	See [a]	See Table 4
Path_Status_A_to_B	See a)	array of Unsigned32	See [a]	See Table 4
Path_Status_B_to_B	See a)	array of Unsigned32	See [a]	See Table 4
Sequence_Number	See a)	Unsigned32	4	See 8.4.1.18
Ethernet DLL trailer				
FCS	See a)		4	CRC based frame check sequence

[a] The field size and offset depends on the number of adapter statuses, the size of each Path_Status field is 4 x number of adapter statuses, the offset is incremented by the size of the previous field.

8.6 CRP Encoding of the AnnunciationFrame

An AnnunciationFrame shall have the format shown in Table 8.

Table 8 – CRP AnnunciationFrame

Parameter name	Offset	Data type	Octet length	Description
Preamble	0		8	Alternating ones and zeros
Destination_Address	8		6	Broadcast, multicast MAC address
Source_Address	14		6	Unicast source MAC address of adapter used to send this message.
EtherType	20		2	0x800 for IP datagrams
IP header				
IP Version	0		4	IP Version (4 bit field) = 4
IP header length				Internet Header Length (4 bit field) is the length of the internet header in 32 bit words = 6
IP service				Service (8 bit field) – set all fields = 0 Bits 0-2: precedence. Bit 3: 0 = normal delay, 1 = low delay. Bits 4: 0 = normal throughput, 1 = high throughput. Bits 5: 0 = normal reliability, 1 = high reliability. Bit 6-7: reserved for future use.
IP total length				Length of IP field (16 bit) in bytes, including IP header and data

Parameter name	Offset	Data type	Octet length	Description
IP identifier	4		4	Identifier for fragmented packets. Set = 0
IP flags				Flags (3 bit field) Bit 0: reserved, set to zero Bit 1: don't fragment = 1 Bit 2: 0 = last fragment
IP fragment offset				Position in original datagram. First fragment = 0
IP time to live	8		4	8 bits. Set to 1
IP protocol				8 bits. User datagram = 17 (decimal)
IP header checksum				16 bits. The checksum field is the 16 bit one's complement of the one's complement sum of all 16 bit words in the header. For purposes of computing the checksum, the value of the checksum field is zero.
IP source address	12		4	Source IP address of this node.
IP destination address	16		4	see 8.4.1.7.
IP options	20		1	IP option field
IP pad	21		3	Pad header to a 32 bit boundary. Pad set = 0
UDP header				
UDP source port	0		2	Annunciation port see 8.4.1.8
UDP destination port	2		2	Annunciation port see 8.4.1.9
UDP length	4		2	Length of UDP field
UDP checksum	6		2	Checksum is the 16-bit one's complement of the one's complement sum of a pseudo header of information from the IP header, the UDP header, and the data.
Body				
CRP LRE_State	0	Unsigned8	1	
CRP Reserved0	1	Unsigned8	1	Reserved
CRP Reserved1	2	Unsigned8	1	Reserved
CRP Duplicate_Detection_State	3	Unsigned8	1	see 8.4.1.16
CRP Node_Index	4	Unsigned16	2	see 8.4.1.11
CRP Max_Node_Index	6	Unsigned16	2	see 8.4.1.12
CRP Operational_IP_Address	8	OctetString	16	see 8.4.1.15
CRP ReservedString	24	VisibleString	32	Reserved
CRP Node_Name	56	VisibleString	32	see 8.4.1.13.
CRP Annunciation_Interval	88	Unsigned32	4	see 8.4.1.14
CRP DiagnosticFrame_UDP_Port	92	Unsigned16	2	see 8.4.1.8 and 8.4.1.9
CRP Reserved	94	Unsigned16	2	Reserved, set to 0.
CRP Configuration_Version	96	Unsigned32	4	see 8.4.1.1
CRP User_Option	100	Unsigned32	4	Reserved
CRP Number_Of_Entries	104	Unsigned32	4	Indicates the number of user-defined entries

Parameter name	Offset	Data type	Octet length	Description
CRP Entries	108	Unsigned32	4 x Number of entries	Reserved for user-defined entries
Ethernet DLL trailer				
FCS	0		4	CRC based frame check sequence

8.7 CRP common protocol

8.7.1 AnnunciationFrames

8.7.1.1 Sending

A node shall send an AnnunciationFrame with the format indicated in Table 8:

- after initial power up or warm start;
- following configuration of the node parameters;
- periodically at a rate determined by the Annunciation_Interval.

A SANC shall send the AnnunciationFrame over its port A.

A DANC shall send the AnnunciationFrame over both its port A and port B.

8.7.1.2 Receiving

The receiving node shall include the node sending the AnnunciationFrame to the Network_Status_Table.

8.7.2 DiagnosticFrames

8.7.2.1 Sending

Nodes shall transmit DiagnosticFrames with the format specified in Table 7:

– at startup, or warm start provided the node is configured.
– following a successful node configuration;
– otherwise, at the rate specified by the DiagnosticFrame_Interval (8.4.1.5) attribute;
– as long as the node is in an operational state.

A SANC shall send a DiagnosticFrame over its port A, with:

– the value of the number of ports field set to 1;
– the Sequence_Number incremented by 1 for each successive message sent;
– the sending adapter set to “1” (LAN_A).

A DANC shall send a pair of DiagnosticFrame over both its port A and port B, with:

– the value of the number of ports field set to 2;
– the Sequence_Number incremented by 1 for each successive sending of a pair, and identical Sequence_Numbers fields in both messages of a pair;
– associated MAC address set;
– the sending adapter set to the value corresponding to the adapter over which the message is sent;
– a time spacing not greater than 30 % of the configured DiagnosticFrame_Interval attribute between the two messages of a pair.

8.7.2.2 Receiving

The node shall be configured to receive DiagnosticFrames on its port using the address specified by the DiagnosticFrame_Address attribute at the DiagnosticFrame_UDP_Destination_Port, see 8.4.1.9.

8.7.3 Detection of duplicate Node_Index

If a node receives a DiagnosticFrame from another node that has the same Node_Index as its own, the node shall set the Duplicate_Detection_State attribute to indicate a duplicate Node_Index was detected.

This entry shall be cleared by reconfiguration of the node with a non-duplicate Node_Index.

8.7.4 Detection of duplicate Node_Name

If a node receives a DiagnosticFrame from another node that has the same Node_Name as its own, the node shall set the Duplicate_Detection_State attribute to indicate a duplicate name detected.

This entry shall be cleared by reconfiguration of the node with a non-duplicate Node_Name.

8.7.5 Failure detection based on arrival of DiagnosticFrames

DiagnosticFrames received by a node shall be used in the construction of the Network_Status_Table unless:

- they indicate duplicate Node_Index detected,
- they have a Node_Index greater than Max_Node_Index, or
- their Node_Index is the same as that for this node.

The node shall track the messages arriving from each reporting node by their Sequence_Number.

The DiagnosticFrame_Interval shall not be considered when evaluating arriving messages.

Arrival of the message shall mark the receive status entry for that reporting node and path that did not receive the message as OK.

If a DANC receives a DiagnosticFrame with a Sequence_Number that exceeds by Max_Sequence_Number_Difference the last DiagnosticFrame received on any other path from the given reporting node, it shall mark the DiagnosticFrame received entry for that reporting node as not OK.

The status shall be returned to OK following receipt of a DiagnosticFrame from the reporting node on the path for which a failure had been recorded provided that its Sequence_Number is no smaller than Max_Sequence_Number_Difference to the highest number received. This shall be the only condition that causes the status to be reset to OK.

The DANC shall reflect the status array of the reporting node contained in the DiagnosticFrame in its Network_Status_Tables in the entry associated with the Node_Index of the reporting node.

However, when a node does not receive a DiagnosticFrame for a particular Node_Index for a time superior to Aging_Time, the adapter A and adapter B columns in the Network_Status_Table for that Node_Index shall indicate that all paths to that node are not OK.

NOTE In DiagnosticFrames sent by receiving nodes, the list of port status reflects this not OK status for that Node_Index.

When receiving nodes begin receiving DiagnosticFrames from a node with that Node_Index, it shall not update the messages received here sent from reporting node's adapter A and B columns in the Network_Status_Table for that Node_Index until Max_Sequence_Number_Difference messages are received. This ensures the DiagnosticFrame content is set correctly.

Two entries of this array shall be examined, that corresponding to the receiving node and that corresponding to the reporting node, see 8.7.6.

8.7.6 Status array entries

8.7.6.1 Receiving node entry

The list of DiagnosticFrame received status array shall indicate whether the reporting node has successfully received DiagnosticFrames on each of the four possible paths from this node. This node shall copy this information from the reporting node's DiagnosticFrame into all four of the reported status extracted from the DiagnosticFrame columns of its Network_Status_Table.

8.7.6.2 Reporting node entry

This entry shall indicate that the reporting node has determined whether there are errors on any of the four possible paths from this node. If this entry indicates any status is not OK, it shall be used to update the DiagnosticFrame received status of the Network_Status_Table for the reporting node. Entries with a status of OK shall not be used.

NOTE This may record the fault more quickly than waiting for detection based on the dual message approach.

8.7.7 Other failure detection

Nodes should provide for the option to use vendor-specific diagnostics or commonly used mechanisms to update their own row in the Network_Status_Table.

The faults detected this way should be sent with the DiagnosticFrame and should allow propagation of the detected fault. Whether or not a node detects errors in its adapters, it shall ensure that its own row is correctly initialized and maintained with an OK value.

8.8 CRP operational messages

8.8.1 Load balancing

Nodes that participate in the CRP redundancy approach are able to balance the load for unicast destination addresses between the available transmission adapters. The Redundancy_Flag [LoadBalancingEnabled] shall control the use of load balancing. 8.8.3 defines how the port is selected when load balancing is used.

8.8.2 LAN and port maintenance

Maintenance may be performed on LAN components or port with reduced impact on system operation, if the port or LAN components are not being used. To avoid use of the port or LAN components, the Redundancy_Flag LoadBalancingEnabled shall be set to false.

NOTE Setting this flag to false has the effect of reverting to the method of using the path associated with this node's operational IP address, or best path if there is a fault.

8.8.3 Selecting transmission path

8.8.3.1 General

Selected transmission paths indicate the ports used to transmit operational messages to specific destination addresses. A path shall be defined as the combination of the transmitting node's sending port and the destination address for the receiving node. Path selection shall be evaluated for a specific unicast or multicast destination address at initial use, and then shall be determined following the detection of a fault or the detection of a repaired fault as described in 8.8.3.2 and 8.8.3.3.

8.8.3.2 Unicast destination address

Path for unicast destination addresses shall be selected according to Table 9.

Table 9 – CRP unicast destination address handling

Destination address of:	Path selection
CRP redundancy participating node with DiagnosticFrame received fields and reported status extracted from DiagnosticFrame fields in Network_Status_Table all OK in the A to A and B to B columns. LoadBalancingEnabled is true	Select path randomly (see [a]) and fairly (equal probability of selection) between the adapters
CRP redundancy participating node with DiagnosticFrame received fields and reported status extracted from DiagnosticFrame fields in Network_Status_Table all OK. LoadBalancingEnabled is false.	Select path that is associated with this node's operational IP address that was configured.
CRP redundancy participating node with DiagnosticFrame received fields and reported status extracted from DiagnosticFrame fields in Network_Status_Table that have one or more not OK.	Select path with no fault
CRP redundancy participating node with Number_of_ports = 1 (singly connected node).	Select path that leads to available adapter
Node that does not have a row in the Network_Status_Table, (it is not participating in CRP redundancy).	Use existing path
Nodes that are outside the network, nodes reachable through routers	Use existing path
[a] "randomly" means that the node selects for a specific path one adapter rather than the other by using a true random number generator, or a pseudo-random number generator with a different seed, each time the node initializes.	

8.8.3.3 Multicast destination address

8.8.3.3.1 Single multicast message transmission adapter enabled state is true

The single_multicast_message_transmission attribute applies to operational messages.

If there are no errors, the node should send on the adapter that has the most singly connected nodes recorded in the Network_Status_Table. This should be selected by choosing the adapter that has the least number of not applicable entries in its DiagnosticFrame received columns of the Network_Status_Table.

If there is a single error, the node shall send on the adapter that has no error recorded in its DiagnosticFrame received or nodes reporting problems columns of the Network_Status_Table.

Otherwise if there are multiple errors, the node shall send on the adapter that has fewest errors recorded in its DiagnosticFrame received and nodes reporting problems columns of the Network_Status_Table.

If the number of errors is the same for each adapter, no change should be made to the selected transmission adapter for multicast addresses.

8.8.3.3.2 Single_multicast_message_reception_adapter_enabled state is false

If all SANs and DANCs can be reached by sending on a single adapter the node shall send on that adapter. That is, if all the “not OK” and “not applicable” entries recorded in its DiagnosticFrame received and nodes reporting problems columns of the Network_Status_Table are associated with the other port.

Otherwise, the node shall send the multicast over both ports.

8.8.4 Selecting reception adapter

NOTE The intention is to reduce duplicate operational multicast messages.

Nodes shall select a reception adapter when the Redundancy_Flags [single_multicast_message_reception_adapter_enabled] is true.

If [single_multicast_message_reception_adapter_enabled] is false, the node shall listen on both adapters. If it is true, the requirements shall be as follows.

If: no LAN faults are detected, the node listens on its port A. (SANs or SANCs only have an Adapter A.)

Else: use same format as above. If a single LAN fault is detected indicating that one adapter of this node cannot correctly receive transmissions, use other adapter of this node.

If there are multiple faults associated with both adapters, listen on both adapters.

8.8.5 Crossed_cable_status

DiagnosticFrames include a field that identifies the port (A or B) on which the message was sent. For certain LAN topologies, this can be used to determine if the port is connected to the LAN correctly.

8.8.6 Configured parameters

Table 10 contains the minimum parameters that are necessary for the protocol. Additional parameters may be added at the user option.

Table 10 – CRP configuration parameters

Parameter	Description	Data type
Device_Id	descriptive string of the end node	OctetString32
Node_Name	end node name	OctetString32
Node_Index	end node’s unique device index	Unsigned16
DiagnosticFrame_UDP_destination_port	UDP port used to receive CRP redundancy messages	Unsigned16
Repeat_Time	end node’s annunciation message Repeat_Time.	Unsigned32
Max_Node_ Index	highest device index used in the CRP network	Unsigned16
Operational_IP_Address	end node’s operational IP address	OctetString16

8.9 CRP services

8.9.1 Configuration options and services

A node may obtain its configuration in three optional ways.

The first is through a local configuration page for those nodes that have a graphical user interface, such as a PC.

The second is through switch configuration where a set of switches configures the device index. The operational IP address would be the sum of a base IP address configured in the device firmware plus the device index.

The third is by a network management node using over the network services that commence upon the reception of an annunciation message. In addition, the network management node has the services to gather the statistics kept by each redundancy capable node. The services are:

- Set_Assignment_Info;
- Get_Redundancy_Info;
- Set_Redundancy_Info;
- Get_Redundancy_Statistics.

8.9.2 LAN redundancy service specification

8.9.2.1 Set_Assignment_Information service

This confirmed service shall be used to set the CRP attributes using the service parameters shown in Table 11.

Table 11 – CRP Set_Assignment_Info service parameters

Parameter name	Req	Ind	Rsp	Cnf
Argument	M	M(=)		
Invoke_Id	M	M(=)		
Source_IP_Address		M		
Source_Port	M	M(=)		
Destination_IP_Address	M	M(=)		
FDA_Address	M	M(=)		
Device_Id	M	M(=)		
Node_Name	M	M(=)		
Node_index	C	C(=)		
DiagnosticFrame_UDP_Destination_Port	C	C(=)		
Repeat_Time	C	C(=)		
Clear_Duplicate_Detection_State	C	C(=)		
Max_Node_Index	C	C(=)		
Operational_IP_Address	C	C(=)		
Result (+)			S	S(=)
Invoke_Id			M	M(=)
Source_IP_Address				M
Destination_IP_Address			M	M(=)
Destination_Port			M	M(=)

Parameter name	Req	Ind	Rsp	Cnf
Repeat_Time			C	C(=)
Max_Node_index			C	C(=)
Result (-)			S	S(=)
Invoke_Id			M	M(=)
Source_IP_Address				M
Destination_IP_Address			M	M(=)
Destination_Port			M	M(=)
Error_Info			M	M(=)
NOTE For the meaning of Req, Ind, Rsp, Cnf, M, U and S, refer to ISO/IEC 10164-1.				

Argument

The argument conveys the parameters of the service request.

Invoke_Id
This parameter contains a value that is determined by the generator of the request and is matched by the responder.
(Unsigned32)

Source_IP_Address
This parameter is the IP address from which the service request was sent. The responder uses it when returning the response.
(OctetString16)

Source_Port
This parameter is the UDP port from which the service request was sent. The responder uses it when returning the response.
(Unsigned16)

Destination_IP_Address
This parameter is the IP address to which the service request is to be sent. The responder uses it when returning the response.
(OctetString16)

FDA_Address
This parameter contains the address to which the service request is being sent.
(Unsigned32)

Device_Id
This parameter contains a descriptive string of the end node.
(OctetString32)

Node_Name
This parameter contains the value of the end node name. The value is not permitted to be blank.
(OctetString32)

Node_index
This conditional parameter contains the value of the end node’s unique device index. Its value is not permitted to be zero.
(Unsigned16)

DiagnosticFrame_UDP_Destination_Port
This conditional parameter contains the value of the UDP port used to receive CRP redundancy messages.
(Unsigned16)

Repeat_Time
This conditional negotiable parameter contains the value of the end node's annunciation message repeat time.
(Unsigned32)

Clear_Duplicate_Detection_State
This conditional parameter causes the duplicate detection state to be set to no duplicates detected if it contains a non-zero value.
(Unsigned8)

Max_Node_Index
This conditional negotiable parameter contains the value of the highest device index used in the CRP network.
(Unsigned16)

Operational_IP_Address
This conditional parameter contains the end node's operational IP address.
(OctetString16)

Result (+)

This parameter indicates that the service request succeeded. The following fields are included in the response. Negotiable parameters may be different than the ones sent by the network manager.

Invoke_Id
Source_IP_Address
Destination_IP_Address
Destination_Port
Repeat_Time
Max_Node_Index

Result (-)

This parameter indicates that the service request failed. The following fields are included in the response.

Invoke_Id
Source_IP_Address
Destination_IP_Address
Destination_Port
Error_Info
This parameter specifies the error condition.

8.9.2.2 Get_Redundancy_Info service

This confirmed service shall be used to retrieve CRP Redundancy attributes using the service parameters shown in Table 12.

Table 12 – CRP Get_Redundancy_Info service

Parameter name	Req	Ind	Rsp	Cnf
Argument	M	M(=)		
Invoke_Id	M	M(=)		
Source_IP Address		M		
Destination_IP_Address	M	M(=)		
Result (+)			S	S(=)
Invoke_Id			M	M(=)
Source_IP_Address				M
Destination_IP_Address			M	M(=)

Parameter name	Req	Ind	Rsp	Cnf
LAN_Redundancy_Attributes_Version			M	M(=)
Number_Of_Network_Adapters			M	M(=)
Max_Message_Number_Difference			M	M(=)
Redundancy_Flags			M	M(=)
Diagnostic_Message_Interval			M	M(=)
Aging_Time			M	M(=)
DiagnosticFrame_Send_Adapter_Address			M	M(=)
DiagnosticFrame_Receive_Adapter_Address			M	M(=)
Result (-)			S	S(=)
Invoke_Id			M	M(=)
Source_IP_Address				M
Destination_IP_Aaddress			M	M(=)
Error_Info			M	M(=)

Argument

The argument conveys the parameters of the service request:

Invoke_Id
Source_IP_Address
Destination_IP_Address

Result(+)

This selection type parameter indicates that the service request succeeded. The following fields are included in the response.

Invoke_Id
Source_IP_Address
Destination_IP_Address
LAN_Redundancy_Attributes_Version
This attribute specifies the version of this object. Each time the value of any of its other attributes changes, the version number is incremented by 1. Version number 0 indicates that this object has not been configured.
(Unsigned32)

Number_Of_Network_Adapters
This attribute specifies the number of adapters on this device. A device may have one or two adapters. They are labelled adapter A and adapter B (adapter B is only used if there are two adapters).
(Unsigned8)

Max_Message_Number_Difference
This attribute defines the maximum acceptable difference between the message number parameters in a pair of diagnostic message service indications received from a single sending device. When the difference exceeds the value of this attribute, a fault in the path from the adapter sending the lower message number is detected.
(Unsigned8)

Redundancy_Flags
This attribute is a bit array that controls how messages are sent and received.
(Unsigned8)

Diagnostic_Message_Interval
This attribute defines the time interval in milliseconds between successive sending of the DiagnosticFrames.
(Unsigned32)

Aging_Time
This attribute defines the time interval in milliseconds used by the LRE to remove silent nodes from its Network_Status_Table.
(Unsigned32)

DiagnosticFrame_Send_Adapter_Address
This attribute defines the send address of the diagnostic frame.
(OctetString16)

DiagnosticFrame_Receive_Adapter_Address
This attribute defines the receive address of the diagnostic frame.
(OctetString16)

Result(-)

This parameter indicates that the service request failed. The following fields are included in the response.

Invoke_Id
Source_IP_Address
Destination_IP_Address
Error_Info

This parameter specifies the error condition.

8.9.2.3 Set_Redundancy_Info service

This confirmed service shall be used to retrieve redundancy attributes. After updating the attributes contained in the request, the responder shall return the values and the updated Redundancy_Attributes_Version number, using the service parameters shown in Table 13.

Table 13 – CRP Set_Redundancy_Info service

Parameter name	Req	Ind	Rsp	Cnf
Argument	M	M(=)		
Invoke_Id	M	M(=)		
Source_IP_Address		M		
Destination_IP_Address	M	M(=)		
Redundancy_Attributes_Version			M	M(=)
Number_Of_Network_Adapters			M	M(=)
Max_Message_Number_Difference			M	M(=)
Redundancy_Flags			M	M(=)
DiagnosticFrame_Interval			M	M(=)
Aging_Time			M	M(=)
DiagnosticFrame_Adapter_Send_Addresses			M	M(=)
DiagnosticFrame_Adapter_Receive_Addresses			M	M(=)
Result (+)			S	S(=)
Invoke_Id			M	M(=)
Source_IP_Address				M
Destination_IP_Address			M	M(=)
LAN_Redundancy_Attributes_Version			M	M(=)
Number_Of_Network_Adapters			M	M(=)
Max_Sequence_Number_Difference			M	M(=)
LAN_Redundancy_Flags			M	M(=)
DiagnosticFrame_Interval			M	M(=)

Parameter name	Req	Ind	Rsp	Cnf
Aging_Time			M	M(=)
DiagnosticFrame_Adapter_Send_Addresses			M	M(=)
DiagnosticFrame_Adapter_Receive_Addresses			M	M(=)
Result (-)			S	S(=)
Invoke_Id			M	M(=)
Source_IP_Address				M
Destination_IP_Address			M	M(=)
Error_Info			M	M(=)
NOTE For the meaning of Req, Ind, Rsp, Cnf, M, U and S, refer to ISO/IEC 10164-1.				

Argument

The argument conveys the parameters of the service request:

Invoke_Id
Source_IP_Address
Destination_IP_Address
Redundancy_Attributes_Version
Number_Of_Network_Adapters
Max_Message_Number_Difference
Redundancy_Flags
DiagnosticFrame_Interval
Aging_Time
DiagnosticFrame_Adapter_Send_Addresses
DiagnosticFrame_Adapter_Receive_Addresses

Result(+)

This selection type parameter indicates that the service request succeeded. The following fields are included in the response.

Invoke_Id
Source_IP_Address
Destination_IP_Address
LAN_Redundancy_Attributes_Version
Number_Of_Network_Adapters
Max_Message_Number_Difference
LAN_Redundancy_Flags
DiagnosticFrame_Interval
Aging_Time
DiagnosticFrame_Adapter_Send_Addresses
DiagnosticFrame_Adapter_Receive_Addresses

Result(-)

This parameter indicates that the service request failed. The following fields are included in the response:

Invoke_Id
Source_IP_Address
Destination_IP_Address
Error_Info

8.9.2.4 Get_Redundancy_Statistics service

This confirmed service shall be used to retrieve statistics attributes using the service parameters shown in Table 14.

Table 14 – CRP Get_Redundancy_Statistics service

Parameter name	Req	Ind	Rsp	Cnf
Argument	M	M(=)		
Invoke_Id	M	M(=)		
Source_IP_Address		M		
Destination_IP_Address	M	M(=)		
Result (+)			S	S(=)
Invoke_Id			M	M(=)
Source_IP_Address				M
Destination_IP_Address			M	M(=)
Number_Of_Diagnostic_Message_Service_Indications_Received			M	M(=)
Number_Of_Diagnostic_Message_Service_Indications_Missed			M	M(=)
Number_Of_Faults_Detected			M	M(=)
List_Of_Crossed_Cable_Status			M	M(=)
Result (-)			S	S(=)
Invoke_Id			M	M(=)
Source_IP_Address				M
Destination_IP_Address			M	M(=)
Error_Info			M	M(=)
NOTE For the meaning of Req, Ind, Rsp, Cnf, M, U and S, refer to ISO/IEC 10164-1.				

Argument

The argument conveys the parameters of the service request:

Invoke_Id
Source_IP_Address
Destination_IP_Address

Result(+)

This selection type parameter indicates that the service request succeeded. The following fields are included in the response.

Invoke_Id
Source_IP_Address
Destination_IP_Address
Number_Of_Diagnostic_Message_Service_Indications_Received
This attribute counts the number of diagnostics messages received from all devices. (Unsigned32)

Number_Of_Diagnostic_Message_Service_Indications_Missed
This attribute counts the number of diagnostics message number gaps from all devices. The detection of each missed message number from any device causes this counter to be incremented.
(Unsigned32)

Number_Of_Faults_Detected
This attribute counts the total number of faults detected from missed diagnostics messages from all devices. The detection of a fault from any device causes this counter to be incremented.
(Unsigned32)

List_Of_Crossed_Cable_Status
This attribute contains the list of crossed cable status. Value 0 means that the list is not present.
(Unsigned32)

Result(-)

This parameter indicates that the service request failed. The following fields are included in the response.

Invoke_Id
Source_IP_Address
Destination_IP_Address
Error_Info

9 CRP Management Information Base (MIB)

```
-- ************************************************************************

IEC-62439-4-MIB DEFINITIONS ::= BEGIN

-- ************************************************************************
-- Imports
-- ************************************************************************

IMPORTS

   OBJECT-TYPE, Counter32,
   TimeTicks, Integer32 FROM SNMPv2-SMI
   Boolean      FROM HOST-RESOURCES-MIB
   MacAddress     FROM BRIDGE-MIB
   iso       FROM RFC1155-SMI;


-- ************************************************************************
-- Root OID
-- ************************************************************************

iec  OBJECT IDENTIFIER ::= { iso 0 }

iec62439-4 MODULE-IDENTITY
   LAST-UPDATED "200811100000Z" -- November 10, 2008
   ORGANIZATION "IEC/SC 65C"
   CONTACT-INFO ""

   DESCRIPTION "This MIB module defines the Network Management interfaces
       for the Redundancy Protocol defined by the IEC 62439 series"

   REVISION  "200612160000Z" -- December 16, 2006
   DESCRIPTION "Initial version of the Network Management interface for the
       Crossnetwork Redundancy Protocol"

   REVISION    "200811100000Z"  --  November 10, 2008
   DESCRIPTION  "
       Separation of IEC 62439 into a suite of documents
       This MIB applies to IEC 62439-4, no change in functionality
              "

   ::= { IEC 62439 }


-- ************************************************************************
-- Redundancy Protocols
-- ************************************************************************

mrp                 OBJECT IDENTIFIER ::= { iec62439 1 }
```

```
prp                 OBJECT IDENTIFIER ::= { iec62439 2 }
crp                 OBJECT IDENTIFIER ::= { iec62439 3 }
brp                 OBJECT IDENTIFIER ::= { iec62439 4 }
drp                 OBJECT IDENTIFIER ::= { iec62439 5 }

-- ***************************************************************************
-- Objects of the CRP Network Management
-- ***************************************************************************

InvokeID OBJECT-TYPE
   SYNTAX Counter32
   MAX-ACCESS read-write
   STATUS  mandatory
   DESCRIPTION
    "Value determined by the requestor matched by the responder"
   ::= { crp 1 }
SourceIPAddress OBJECT-TYPE
   SYNTAX OCTET STRING (SIZE(1..16))
   MAX-ACCESS read-write
   STATUS  mandatory
   DESCRIPTION
    "the IP address from which the service request was sent"
   ::= { crp 2 }
SourcePort OBJECT-TYPE
   SYNTAX INTEGER
   MAX-ACCESS read-write
   STATUS  mandatory
   DESCRIPTION
    "UDP port from which the service request was sent"
   ::= { crp 3 }
DestinationIPAddress OBJECT-TYPE
   SYNTAX OCTET STRING (SIZE(1..16))
   MAX-ACCESS read-write
   STATUS  mandatory
   DESCRIPTION
    "the IP address to which the service request is to be sent"
   ::= { crp 4 }
DeviceID OBJECT-TYPE
   SYNTAX OCTET STRING (SIZE(1..32))
   MAX-ACCESS read-write
   STATUS  mandatory
   DESCRIPTION
    "a descriptive string of the end node"
   ::= { crp 5 }
NodeName OBJECT-TYPE
   SYNTAX OCTET STRING (SIZE(1..32))
   MAX-ACCESS read-write
   STATUS  mandatory
   DESCRIPTION
    "end node name"
   ::= { crp 6 }
NodeIndex OBJECT-TYPE
   SYNTAX INTEGER
   MAX-ACCESS read-write
   STATUS  mandatory
   DESCRIPTION
    "the value of the end node's unique device address"
   ::= { crp 7 }
DiagnosticFrameUDPDestinationPort OBJECT-TYPE
   SYNTAX INTEGER
   MAX-ACCESS read-write
   STATUS  mandatory
   DESCRIPTION
    "the value of the UDP port used to receive FRP redundancy messages"
   ::= { crp 8 }
RepeatTime OBJECT-TYPE
   SYNTAX INTEGER
   MAX-ACCESS read-write
   STATUS  mandatory
   DESCRIPTION
    "the value of the end node's annunciation message repeat time"
   ::= { crp 9 }
ClearDuplicateDetectionState OBJECT-TYPE
   SYNTAX INTEGER {
    noOp (0),
    clearDuplicateDetectionState (1)
    }
   MAX-ACCESS write-only
```

```
   STATUS  mandatory
   DESCRIPTION
    "causes the duplicate detection state to be set to no duplicates detected if it
contains a non-zero value"
   ::= { crp 10 }
MaxNodeIndex OBJECT-TYPE
   SYNTAX INTEGER
   MAX-ACCESS read-write
   STATUS  mandatory
   DESCRIPTION
    "the value of the highest device index used in the CRP network"
   ::= { crp 11 }
OperationalIPAddress OBJECT-TYPE
   SYNTAX OCTET STRING (SIZE(1..16))
   MAX-ACCESS read-write
   STATUS  mandatory
   DESCRIPTION
    "the end node’s operational IP address"
   ::= { crp 12 }

END
```

Bibliography

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IEC 62439-5, *Industrial communication networks – High availability automation networks – Part 5: Beacon Redundancy Protocol (BRP)*

IEC 62439-6, *Industrial communication networks – High availability automation networks – Part 6: Distributed Redundancy Protocol (DRP)*

ISO/IEC 10164-1, *Information technology – Open Systems Interconnection – Systems Management : Object Management Function*

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